

Yaxin Xiao

Machine Learning Privacy and Security Researcher

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research profile

Researcher focused on privacy and security for machine learning, especially model extraction, membership inference, machine unlearning, and watermarking for model IP protection.

Current work is centered at The Hong Kong Polytechnic University in the Department of Electrical and Electronic Engineering.

Education

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| PhD The Hong Kong Polytechnic University , Electrical and Electronic Engineering <ul style="list-style-type: none"> • Advisor: Prof. Haibo Hu • Supported by the Hong Kong PhD Fellowship Scheme (HKPFS) | Hong Kong SAR
Sept 2020 – Sept 2025 |
| MEng Zhejiang University , Engineering <ul style="list-style-type: none"> • Research training in computer and information engineering | Hangzhou, China
Sept 2017 – June 2020 |
| BEng Zhejiang University , Engineering <ul style="list-style-type: none"> • Strong foundation in engineering and applied research | Hangzhou, China
Sept 2013 – June 2017 |

Experience

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| The Hong Kong Polytechnic University , Postdoctoral Fellow / Researcher <ul style="list-style-type: none"> • Research on model extraction and membership inference attacks • Research on machine unlearning and watermarking for model protection | Hong Kong SAR
Sept 2025 – present
10 months |
| The Hong Kong Polytechnic University , Teaching Assistant / Laboratory Instructor <ul style="list-style-type: none"> • EIE4432 Web Systems and Technologies • EIE4118 Intrusion Detection and Penetration Testing • EIE4117 Capstone Projects and EIE3117 Integrated Project | Hong Kong SAR
Sept 2020 – June 2025
4 years 10 months |

Publications

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| MExMI: Pool-based Active Model Extraction Crossover Membership Inference
Pool-based active extraction attack that connects model stealing and membership inference.
<i>Yaxin Xiao, Qingqing Ye, Haibo Hu, Huadi Zheng, Chengfang Fang, Jie Shi</i>
(NeurIPS 2022) | Dec 2022 |
| Privacy Attacks Exploiting Residuals: How Approximate Machine Unlearning Fails to Protect Privacy
Demonstrates how approximate unlearning can still leak membership information through residual traces.
<i>Yaxin Xiao, Qingqing Ye, Li Hu, Huadi Zheng, Haibo Hu, Zi Liang, Yijie Jiao</i>
(ICCV 2025) | Oct 2025 |
| Unlocking High-Fidelity Learning: Towards Neuron-Grained Model Extraction
Explores neuron-grained extraction for high-fidelity recovery of victim model behavior.
<i>Yaxin Xiao, Haibo Hu, Qingqing Ye, Li Tang, Zi Liang, Huadi Zheng</i>
(IEEE Transactions on Dependable and Secure Computing, 2025) | 2025 |

Attacks

A black-box watermarking scheme designed to withstand model extraction attacks.

Yaxin Xiao, Qingqing Ye, Zi Liang, Haoyang Li, RongHua Li, Huadi Zheng, Haibo Hu
(AAAI 2026)

Selected Honors

- Hong Kong PhD Fellowship Scheme (2020)
- Outstanding Graduate Leader Award of Zhejiang University (2018)
- Zhejiang University Award of Honor for Graduate (2018)
- Outstanding Graduates of Zhejiang University (2017)
- National Scholarship of Chinese Government (2016)

Skills

Research Areas: Model extraction, membership inference, machine unlearning, watermarking, ML privacy, ML security

Academic Service: Reviewer for ICCV, ICLR, WWW, CVPR, ICML, and IEEE TIFS; reviewer under supervision for ICDE and VLDB

Teaching: Web systems, intrusion detection, capstone supervision, and laboratory instruction